

Amirtheshwaran "Amirthesh" Navaneethakrishnan

Charlotte, NC | +1 704-648-6035 | amirthesh14@gmail.com

linkedin.com/in/amirtheshn | github.com/Amirtheshwaran

EDUCATION

University of North Carolina at Charlotte

Master of Science in Computer Science, GPA 3.5/4.0

Charlotte, NC

Expected May 2027

SRM Institute of Science and Technology

Bachelor of Technology, Computer Science and Engineering (AI and ML)

Kattankulathur, India

2021 - 2025

EXPERIENCE

Drata IT Solutions

Remote | Jul 2024 - Aug 2026

Cloud Infrastructure Intern

Jun 2026 - Aug 2026

- Deployed 8 containerized services across Compute Engine, GKE, Cloud Run, and Cloud SQL, packaging TypeScript and Bun applications as Docker images and cutting image size 35%.
- Provisioned VPCs, subnets, load balancers, firewall rules, and DNS for 3 environments in 2,000+ lines of Terraform, replacing a 2-day manual process with a repeatable 30-minute apply.
- Built the build, test, and deploy pipeline that took deploys from 25 minutes to 10, and instrumented every service with centralized logging, monitoring, and audit trails.

Software Engineer Intern, Backend

Jul 2024 - Nov 2024

- Owned 12 Python REST endpoints end to end, from schema design through integration with 3 third-party services, testing, and production deployment.
- Defined API contracts and payload schemas jointly with 4 frontend engineers, removing the integration mismatches that had been blocking releases.
- Cut average SQL query time 40%, refactored 6 backend modules into shared components, and reviewed 20+ pull requests.

University of North Carolina at Charlotte

Charlotte, NC

Teaching Assistant, Frontend Web Development (ITIS 3135)

Aug 2026 - Present

- Hold office hours and grade labs and assignments for 118 students building sites with HTML, CSS, and JavaScript, working through their code to find and explain bugs.

PROJECTS

Gait-Based Person Re-Identification Pipeline

Jan 2026 - Apr 2026

Python, YOLOX, ByteTrack, OpenGait | github.com/PritomKumarPaul/gait_tracking

- One of 4 developers on a 5-stage computer vision pipeline covering detection, multi-object tracking, silhouette extraction, gait embedding, and ranking, reaching 85% rank-1 accuracy across gallery and probe sets.
- Traced model initialization failures to conflicting dependency versions, resolved 10+ conflicts, and restructured the dataset layout so results reproduce across runs.

Dental Demand Platform

Jan 2026 - Apr 2026

Python, Streamlit, Scikit-learn | dentaldemandplatformteam19.streamlit.app

- Sole developer (one-person course team) of a deployed platform that ranks dental service demand across 3,000+ US counties, ingesting and joining county-level CDC PLACES public health data.
- Built the scoring layer over 10+ US Census variables, normalizing indicators onto a common scale and running ML inference to produce a ranked demand estimate per county.

Staff Training Simulator

Aug 2025 - Present

React, Vite, Netlify Functions | amirtheshwaran.github.io/urec-shift-report-simulator

- Built and deployed a serverless training app with 12 randomized scenarios and dynamic forms that scores submitted reports through an LLM-backed endpoint.
- Added a local fallback scoring path so the app stays usable when the model endpoint is unavailable.

AnonU, Anonymous Microblogging App

Dec 2024 - Apr 2025

React Native, Firebase, Firestore | github.com/Amirtheshwaran/anonu-remake

- Shipped a cross-platform iOS and Android app to roughly 100 active users, supporting anonymous feeds, nested comment threads, and real-time sync.
- Designed the Firestore data model and security rules across 8 collections, integrated Firebase Auth, and built 15 reusable UI components with client-side validation.

TECHNICAL SKILLS

Languages: Java, Python, C++, TypeScript, JavaScript, C#, C, SQL

Cloud and Infrastructure: GCP (GKE, Cloud Run, Cloud SQL), AWS, Docker, Kubernetes, Terraform, CI/CD

Frameworks and Tools: React, React Native, Django, Bun, Vite, Firebase, Firestore, MySQL, Scikit-learn, OpenCV, Git